

INDEPENDENT STUDY

— ACADEMIC YEAR ONLY



PERSONAL INFORMATION

Full Name : Vedant Hemant Upadhyay

Student ID # : 2172139

E-Mail : veupadhy@ucsc.edu

Program : CSE

Quarter : WINTER

INDEPENDENT STUDY / THESIS RESEARCH

297 Independent Study/Research
Coursework still in progress
Thesis Research
299 Coursework completed, continuing research

Class Number 43103

Course Number CSE 297B

Section Number 6

****Important** - Please select the **home** department of the Faculty Sponsor

AM 297 ☐ BME 297 ☐ CMPM 297 ☐ CSE 297 ☒ ECE 297 ☐ STAT 297 ☐

AM 299 ☐ BME 299 ☐ CMPM 299 ☐ CSE 299 ☐ ECE 299 ☐ STAT 299 ☐

This individual study is equivalent to 2 ☐ 5 ☒ 10 ☐ 15 ☐ units.

*This form is to be completed, approved, and filed with the Baskin Engineering Graduate Affairs Office (GAO). Class numbers are issued by the GAO **via e-mail**. You must enroll through "MyUCSC" after receiving your class number.*

- 1. Objectives of the proposed work:** **Please be sure you have discussed these objectives with your mentor before filling out this section of the form**

To design and evaluate a Linux-inspired CPU scheduling simulator that compares classical scheduling algorithms with machine-learning-guided policies to study performance and overhead tradeoffs.

- 2. Expectations for this Independent Study:** **Please be sure you have discussed these expectations with your mentor before filling out this section of the form, as they will form the basis for the final evaluation**

The project will result in a working scheduling simulator, experimental evaluation of multiple scheduling strategies, and a written analysis documenting findings and insights on ML-based scheduling effectiveness.

48 HOUR BUSINESS TURNAROUND



**** Important:** for sections 3 & 4, please be sure you have discussed these expectations (**meeting schedule & hours per week**) with your mentor **before** filling out these sections of the form ******

3. Mentor/Mentee tentative meeting schedule:

☐ Bi-Weekly
 ☒ Weekly
 ☐ Bi-Monthly
 ☐ Monthly
 ☐ Other

If Other:

Meetings - (Other) explanation

4. **Estimated number of hours per week:** Independently on project: With Mentor:

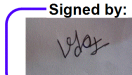
(Hours should match the credits received, with a rough conversion of 3 hours effort for each unit of credit)

SIGNATURES

Vedant Hemant Upadhyay

Student Name

Signed by:



2947CAB076B94B0...

Student Signature

1/26/2026

Date

Scott Brandt

Faculty Name

DocuSigned by:



FB2A99F580784D6...

Faculty Signature

1/26/2026

Date

OPTIONAL COMMENTS FOR FACULTY MENTORS:

****Please add any additional information or modifications here**** (Graduate students will receive any updates stated in this section of the form)



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veupadhy@ucsc.edu

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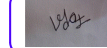
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ID: f20d97fc-46f8-49fc-9788-2d5fb1c16a2b

Scott Brandt

sbrandt@ucsc.edu

Vice Chancellor, Research

University of California, Santa Cruz

Security Level: Email, Account Authentication

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University of California, Santa Cruz

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Status

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Status

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Status

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Required hardware and software

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